SUMMARY REPORT ON KDE PLOTS

Phase 1 — Planning and Header File Design

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Project: Data Visualization Library in C++

Module: Kernel Density Estimation (KDE) Plots

Reporting Period: Week 1

Phase: Phase 1 — Planning and Header File Design

1. Objective

The objective of Phase 1 was to plan the KDE Plots module and create its header files (.hpp) to define the functions and interfaces required for implementing Kernel Density Estimation plots in the C++ Data Visualization Library.

2. Purpose of the Module

The KDE Plots module is intended to visualize the distribution of numerical data using Kernel Density Estimation. KDE estimates the probability density of a dataset by producing a smooth curve that shows where data values are concentrated.

Unlike a histogram, which groups values into separate bins, a KDE plot provides a continuous curve that helps users understand the overall shape and distribution of their data.

3. Work Completed

During Phase 1, the main focus was on planning the module and preparing its header files (.hpp).

- Creating the KDE Plots header file to declare the functions and interfaces needed by the module.
- Planning how the module will receive and process numerical datasets.
- Identifying the need for parameters such as bandwidth, which controls the smoothness of the estimated density curve.
- Considering how the KDE module will connect with the wider Data Visualization Library.
- Establishing a foundation for implementing the KDE calculation and plotting functionality in the next phase.

The header files provide the interface through which other components of the library can access the KDE functionality.

4. Planned Work

Phase 2 — Implementation: Write the C++ source code to calculate KDE values and generate the points needed to draw the density curve.

Phase 3 — Testing: Test the module using different datasets and bandwidth values to verify the accuracy of the density estimates.

Phase 4 — Integration: Connect the KDE module to the main plotting system and ensure it works with other components of the library.

Phase 5 — Documentation: Document the functions, parameters, and usage of the KDE module.

5. Challenges and Considerations

• Choosing a suitable kernel function, such as the Gaussian kernel.

• Determining an appropriate bandwidth to control the smoothness of the curve.

• Handling empty datasets and invalid input values.

• Ensuring that the density calculations are accurate and computationally efficient.

• Designing the header files so that the module can integrate easily with other parts of the library.

6. Conclusion

Phase 1 has focused on planning the KDE Plots module and preparing its header files (.hpp). This work establishes the interface and structure required for the next stage of development. The next step is to implement the KDE calculations, test the resulting density estimates, and integrate the module into the Data Visualization Library.